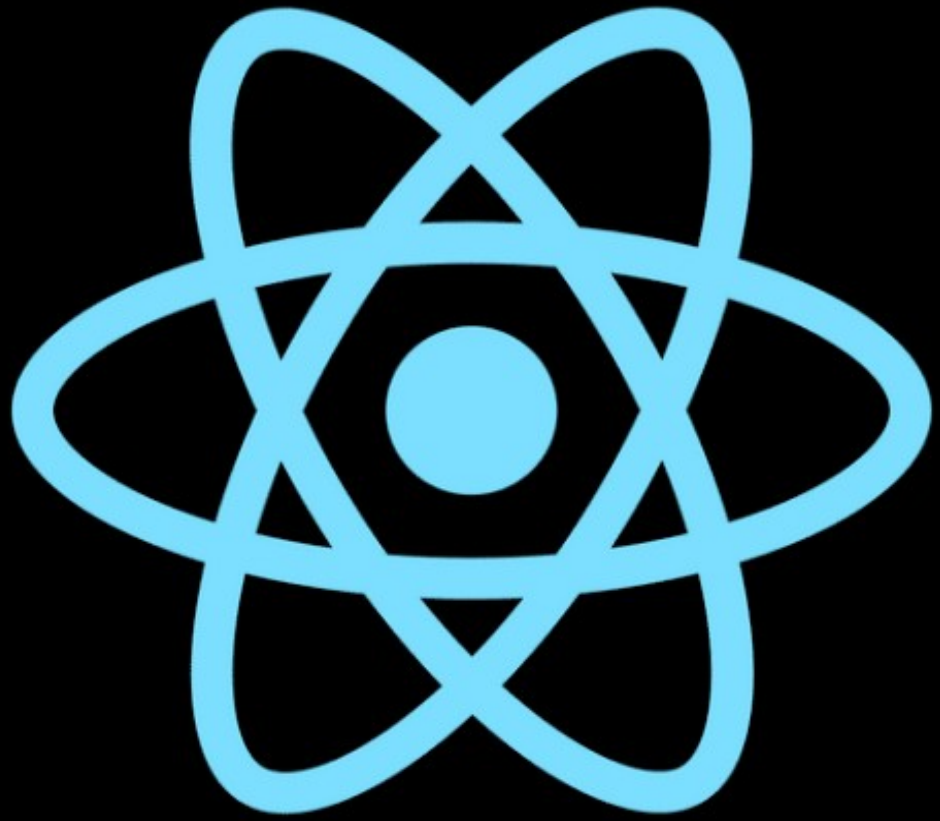


React JS

REACT JS

MERN DAY 1

REACT JS

- It is a JavaScript library for building user interfaces.
- It is the view layer for web applications.
- Developed at Facebook and released to the world in 2013,
- It drives some of the most widely used apps, powering Facebook and Instagram among countless other applications.
- Ideal for building single-page applications (SPAs) and interactive web applications.

Single Page Web Applications

- A Single-Page Application (SPA) is a web application or website that interacts with the user by dynamically rewriting the current page rather than loading entire new pages from the server.
- In SPAs, the initial HTML, CSS, and JavaScript are loaded once, typically during the initial page load, and subsequent interactions with the application are handled by JavaScript, which fetches data from the server and updates the page content without requiring a full page reload.

Key characteristics of SPA

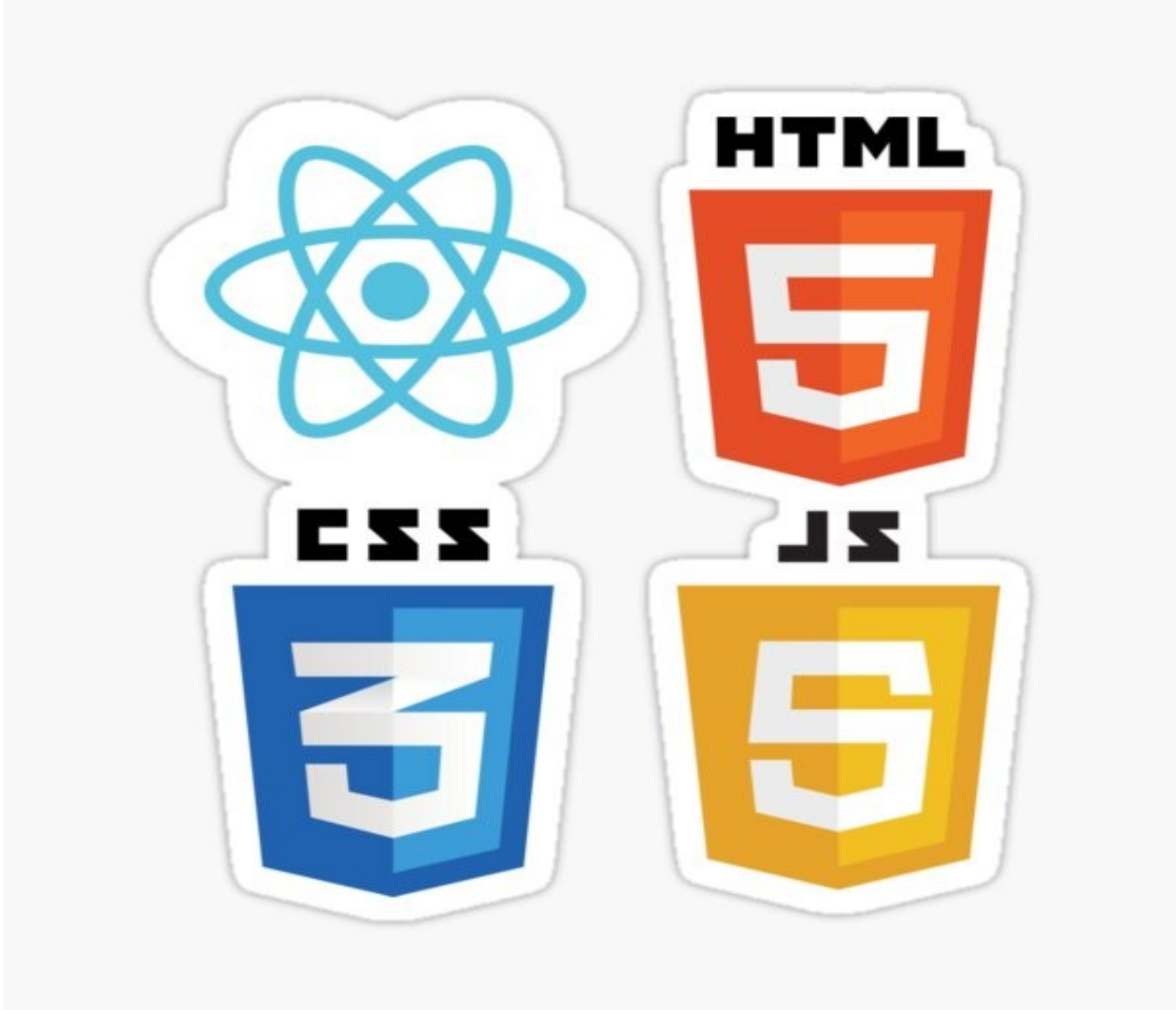
- **No Page Reloads**: SPAs do not trigger full page reloads when a user interacts with the application. Traditional multi-page applications (MPAs) often require the entire page to be refreshed when navigating between different parts of the site.

- **Dynamic Loading**: SPAs load and render content dynamically. When a user clicks on a link or interacts with the interface, the application uses JavaScript to fetch and update only the necessary data and UI components, resulting in a smoother and faster user experience.

- **Client-Side Routing:** SPAs often implement client-side routing using JavaScript libraries or frameworks. This allows for the creation of distinct routes or URLs within the application without requiring server-side changes.

React is a Library?

- library typically refers to a collection of pre-written code, functions, and components that developers can use to build user interfaces (UIs) for web applications.
- React itself is often referred to as a library because it provides a set of tools and components for building UIs efficiently.

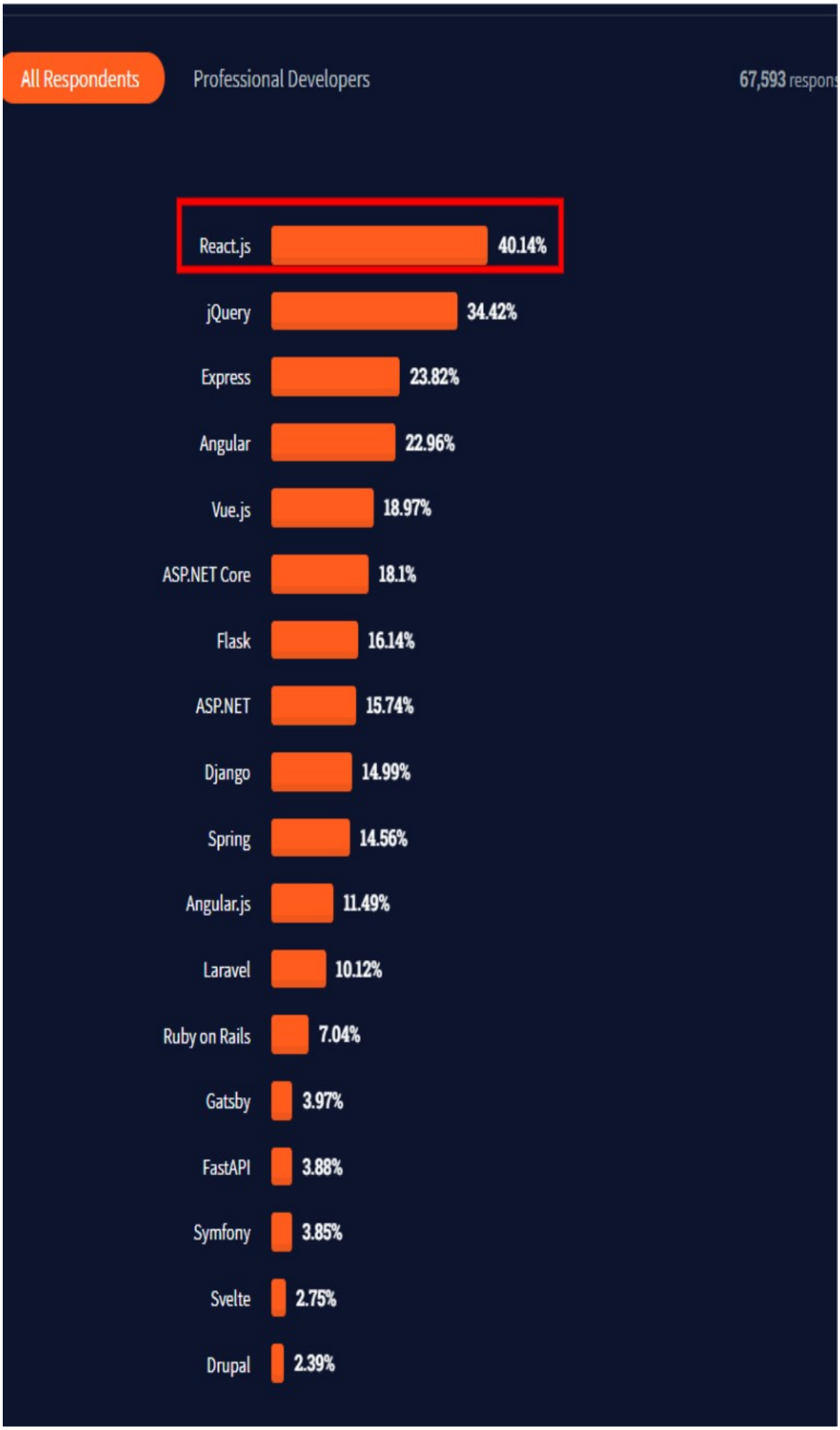

HTML
CSS
JS

Features of REACT JS

- Easy to learn
- Simple
- Native approach
- Data binding
- Performance
- Testability

Which Popular Sites Use ReactJS?

- FACE BOOK
- INSTAGRAM
- NETFLIX
- WHATSAPP
- YAHOO MAIL
- DROP BOX


All Respondents
Professional Developers
67,593 respons
React.js 40.14%
jQuery 34.42%
Express 23.82%
Angular 22.96%
Vue.js 18.97%
ASP.NET Core 18.1%
Flask 16.14%
ASP.NET 15.74%
Django 14.99%
Spring 14.56%
Angular.js 11.49%
Laravel 10.12%
Ruby on Rails 7.04%
Gatsby 3.97%
FastAPI 3.88%
Symfony 3.85%
Svelte 2.75%
Drupal 2.39%

ReactJS — Installation

- React CLI tools depends on the Node.js and must be installed in your system.

```
node –v
```

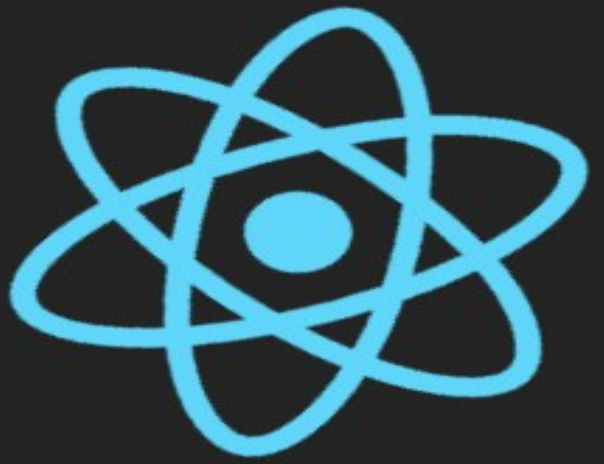
- Create React App is a modern CLI tool to create single page React application.

```
npm install -g create-react-app
create-react-app my-app
cd my-app
npm start
```

- Using npx

```
npx create-react-app myapp
```





Vite + React

Vite Installation

- npm create vite@latest

You can also directly specify the project name and the template you want to use via additional command line options. For example, to scaffold a Vite + React project, run

- npm create vite@latest my-react-app -- --template react

- cd my-project
- npm run dev

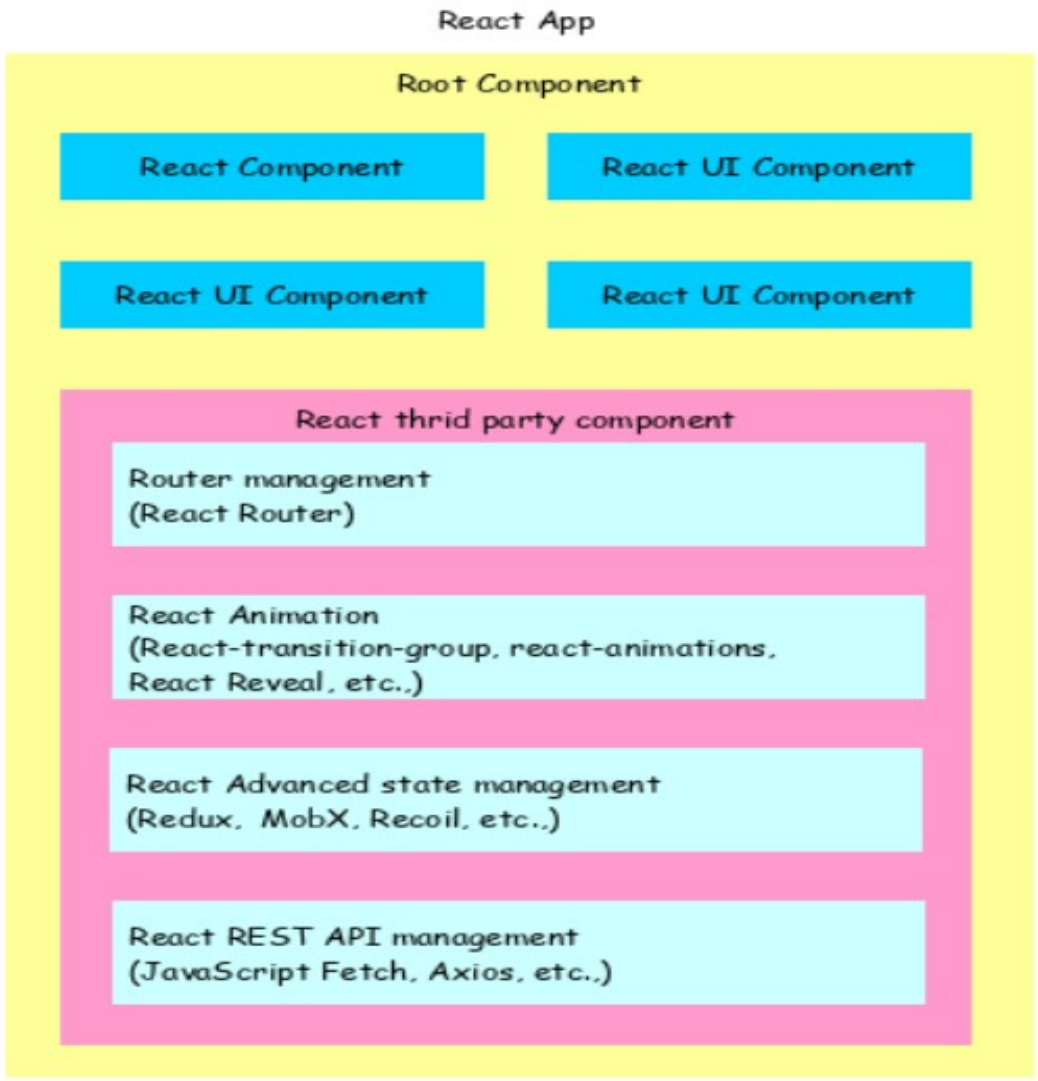
Components

- React app starts with a single root component.
- Root component is build using one or more component.
- Each component can be nested with other component to any level.
- Most of the components are user interface components.
- React app can include third party component for specific purpose such as routing, animation, state management, etc.

Component

- The primary job of a React component is to render its user interface and update it whenever its internal state is changed.
- **React component provides below functionalities**.
  - Initial rendering of the user interface.
  - Management and handling of events.
  - Updating the user interface whenever the internal state is changed.

Understand the high level architecture of a React application



Folder structure

- **node_modules(Folder)**

Contains all the dependencies that are needed for an initial working react app

- **.gitignore(file)**

This file specifies intentionally untracked files that Git should ignore

- **package.json(file)**

This file contains various **metadata** that is relevant to our project.

It specifies the **dependencies** being used in the project which helps npm setup same environment on different machine for our project

- **README.md(file)**

This file can be used to define ***usage, build instructions, summary of project***, etc. It uses markdown markup language to create content.

Public folder and it’s files

- **public(folder)**

Root folder that gets served up as our react app.

- **favicon.ico(file)**

It’s an icon file that is used in index.html as favicon.

- **index.html(file)**

It is the template file which is served up when we run **start script** to launch our app. It is considered best practice not to create multiple html file in public folder instead use this file and inject react components in this file’s root div container. Other css libraries, etc can be defined in this files.

Src folder and its files

- **src(folder)**

In simplest form it’s our react app folder i.e. containing **components, css files** etc. It’s the mind of our app.

- **App.css(file)**

Contains styles of our react component(App.js)

- **App.jsx(file)**

This file has very basic react component defined which can be replaced by our own root component

- **index.css(file)**

Contains styles for general setup of our app.

- **main.jsx(file)**

This files renders our component and registers service workers(unregistered by default)

Task

- What is npm?
- What is the difference between npm and npx?
- What is the difference between framework and library
- What is purpose of using package.json?
- What is React Fragment?

Thank You